Analysis of Algorithms

Project – Proposal

Students enrolled in *Analysis of Algorithms* are expected to work on a project during the semester, which will be 30% of the final grade, split between two different terms. There are two presentations of the project. In the first presentation, the students present the first version of the algorithm, the expectation is to have a complete project including the implementation and the analysis. In the second presentation, the student presents new results, updated optimizations, and addresses any comments from the first version.

What is the project about?

Your job is to become the class's expert on a given algorithm, identify a meaningful problem to which it can be applied, understand its theoretical properties, formulate questions about its behavior, design experiments to answer those questions, compare it against appropriate alternatives, and defend their conclusions. Each person is assigned an algorithm, look at the assignments sheet.

What do I need to include in the proposal?

1. One-page proposal that includes a brief description of the selected algorithm and a description of the problem it solves.
    1. What kinds of problems can this algorithm solve?
    2. What are its assumptions?
    3. What would constitute an interesting application?
    4. What data representation is appropriate?
    5. What competing approach should be considered?
    6. What behavior is worth investigating?
2. Submit as a PDF file.
3. Check the class schedule for information about the deadline.

Instructions:

- We want to see your creativity and how you exploit the tool you are given.
- Usage of AI is expected and encourage for the different parts of the project.
- As you progress in your project always ask yourself: Am I learning and growing?
- Be alert of deadlines.